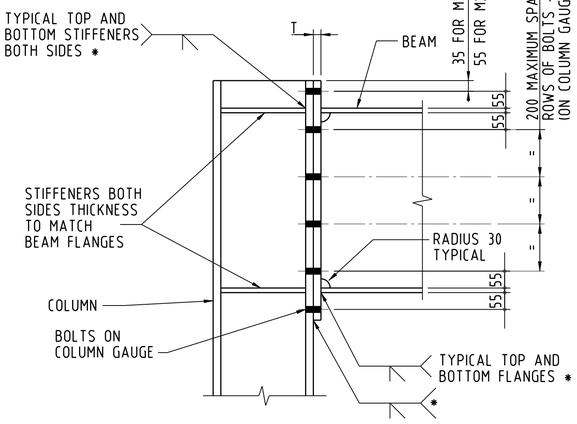
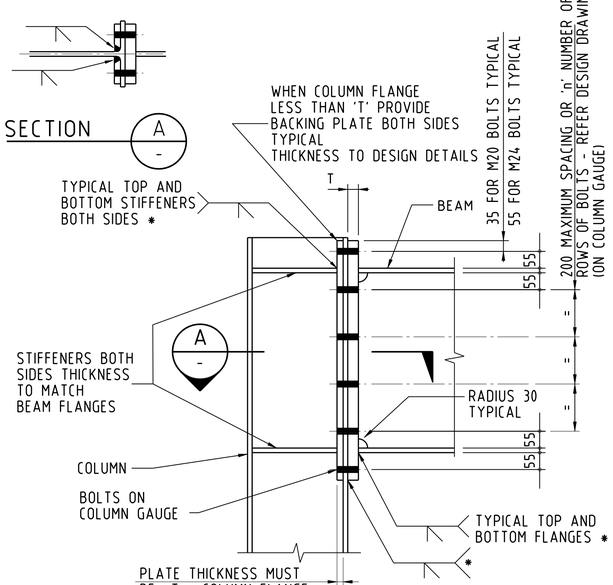


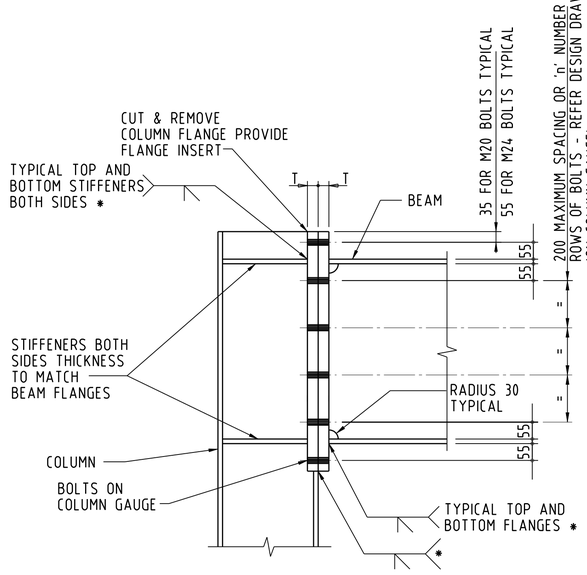
BOLTED TYPE MCA

PREFERRED WHERE 'T' REQUIRED IS EQUAL TO OR LESS THAN COLUMN FLANGE THICKNESS.



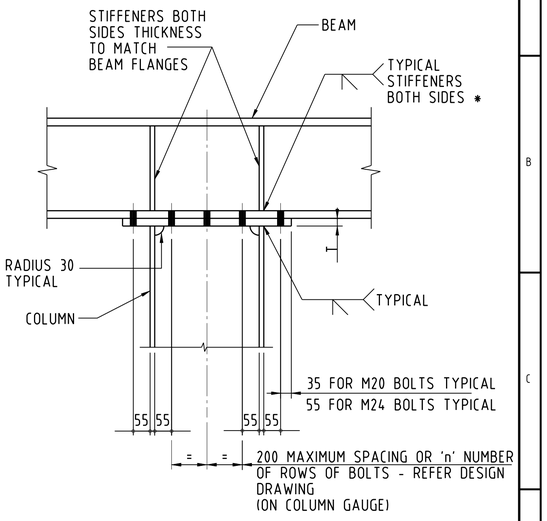
BOLTED TYPE MCB

NON PREFERRED OPTION WHERE 'T' IS GREATER THAN COLUMN FLANGE THICKNESS.



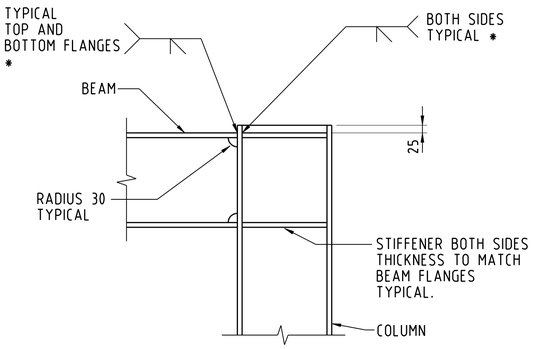
BOLTED TYPE MCC

PREFERRED WHERE 'T' REQUIRED IS GREATER THAN COLUMN FLANGE THICKNESS.

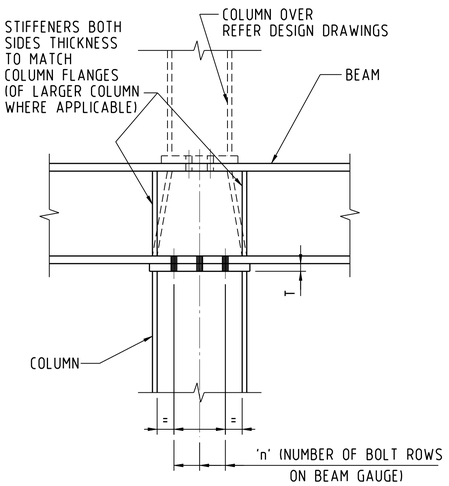


BOLTED TYPE MCD

BOLTED AND WELDED PORTAL FRAME MOMENT CONNECTIONS

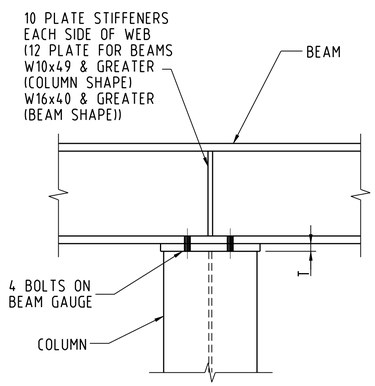


WELDED TYPE MCW



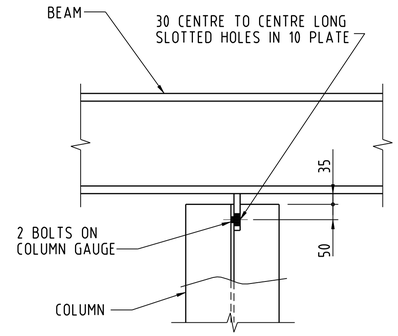
TYPE BCCA

(COLUMN OVER SIMILAR)



TYPE BCCB

(COLUMN OVER SIMILAR)



TYPE BCCC

(COLUMN OVER SIMILAR)

LEGEND AS DENOTED ON DESIGN DRAWINGS

MCA/(n)/(p)/T/D

- D FOLLOWED BY A NUMBER DENOTES BOLT DIAMETER (OTHER THAN M20)
- T FOLLOWED BY A NUMBER DENOTES END PLATE THICKNESS (AND FLANGE INSERT THICKNESS TYPE MCC ONLY)
- DENOTES BOLTING PROCEDURE OTHER THAN SNUG TIGHT (ST)
- DENOTES NUMBER OF BOLT ROWS (WHERE No OF BOLT ROWS IS GREATER THAN MINIMUM.
- DENOTES MOMENT CONNECTION TYPE

LEGEND AS DENOTED ON DESIGN DRAWINGS

BCCA/n/(p)/T/D

- D FOLLOWED BY A NUMBER DENOTES BOLT DIAMETER (OTHER THAN M20)
- T FOLLOWED BY A NUMBER DENOTES END PLATE THICKNESS
- DENOTES BOLTING PROCEDURE OTHER THAN SNUG TIGHT (ST)
- DENOTES NUMBER OF BOLT ROWS
- DENOTES BEAM/COLUMN CONNECTION TYPE

BEAM OVER/UNDER COLUMN CONNECTIONS

NOTES:

1. FOR STEELWORK GENERAL NOTES REFER DRAWING MD-101-SMEM-EG-ST-DST-0009.
2. * DENOTES USE FULLY PREPARED DOUBLE BUTT WELD WHERE PLATE OR FLANGE THICKNESS ⩾ 16
3. FOR COLUMN (BEAM) GAUGE DETAILS REFER TABLE 2 DRAWING MD-101-SMEM-EG-ST-DST-0010.
4. STRUCTURAL CAPACITIES OF THE CONNECTIONS SHOWN ON THIS DRAWING SHALL BE CHECKED BY THE CONTRACTOR TO DETERMINE THEIR SUITABILITY FOR THE INTENDED USE.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD

This drawing and all the information contained herein it is the proprietary property of Ma'aden and is subject to confidentiality restrictions between Ma'aden and the recipient. The information contained herein is provided for a limited purpose and must be returned on request. Ma'aden makes no respresentations or warranties that this information is complete or fit for any particular purpose. Neither this drawing nor any information concerning it may be copied, exhibited or furnished to others. Nor may photographs be taken of any article fabricated or assembled from this drawing, without Ma'aden prior written consent.

REV C01 PREPARED FOR MA'ADEN BY HATCH ASSOCIATES H332701-0000-35-041-0017_C

MA'ADEN معادن Saudi Arabian Mining Company

DRAWN	J BADAL	09.03.11
CHECKED	J EDWARDS	18.04.12
ENGINEER	D KIRKWOOD	18.04.12
APPROVED	R COVELLA	18.04.12
MA'ADEN		
SCALE NTS		A1

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
MOMENT, COL UNDER/OVER CONNECT'S

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0017 REV C01

$DATE$ $TIME$ $FILE$